Develop and Test Custom SIEM Correlation Rules Using ELK Stack

1. Objective

The objective of this task was to configure and test custom detection rules using the ELK Stack to identify suspicious security events, including brute-force login attempts, potential DNS tunneling, and suspicious PowerShell execution.

The task helped me understand how SIEM detection rules can identify suspicious activity from indexed security logs and generate alerts for further investigation.

2. Lab Environment

Component	Details
Operating System	Kali Linux
SIEM Platform	ELK Stack
Log Analysis	Kibana
Search and Storage	Elasticsearch
Log Index	`security-logs`

3. Lab Setup

I installed and configured Elasticsearch and Kibana on Kali Linux. I created the `security-logs` index to store security events and used Kibana to search and analyze the indexed data.

I then configured three custom detection rules for suspicious authentication failures, potentially suspicious DNS queries, and suspicious PowerShell commands.

Log Analysis Workflow

Kali Linux → Elasticsearch → `security-logs` → Kibana Detection Rules → Alerts → Investigation

4. Detection Rule 1: Brute-Force Detection

Rule Configuration

Setting	Value
Rule Name	Brute-Force Detected
Index	`security-logs`
Rule Type	Threshold
KQL Query	`event.category: "authentication" AND event.outcome: "failure"`
Group By	`source.ip.keyword`
Threshold	5 events
Runs Every	5 minutes
Additional Look-back	6 minutes
Severity	High
Risk Score	73
MITRE ATT&CK	T1110 – Brute Force

Purpose

This rule identifies repeated authentication failures that may indicate brute-force activity. It groups matching events by source IP and triggers when the configured threshold is reached.

Testing and Expected Result

Five or more matching authentication failures from the same source IP within the configured time window should meet the threshold.

Repeated failures alone do not prove a successful attack, so the source IP, timestamps, affected account, and any subsequent successful logins should be investigated.

5. Detection Rule 2: Potential DNS Tunneling

Rule Configuration

Setting	Value
Rule Name	DNS Tunneling
Index	`security-logs`

Rule Type	Custom Query
KQL Query	`network.protocol: "dns" AND dns.question.type: ("TXT" OR "NULL" OR "CNAME")`
Runs Every	5 minutes
Additional Look-back	6 minutes
Severity	High
Risk Score	73
MITRE ATT&CK	T1071.004 – DNS

Purpose

This rule searches for DNS events using selected record types that may be relevant during a DNS tunneling investigation.

Testing and Expected Result

I used a controlled DNS test event to validate whether the query matched the indexed data. Matching events can be investigated for unusual query frequency, long or encoded subdomains, suspicious destinations, and unexpected DNS activity.

TXT, NULL, or CNAME records alone do not confirm DNS tunneling. Additional evidence is required to determine whether the activity is malicious.

6. Detection Rule 3: Suspicious PowerShell Execution

Rule Configuration

Setting	Value
Rule Name	Suspicious PowerShell Execution Detected
Index	`security-logs`
Rule Type	Custom Query
KQL Query	`process.name: "powershell.exe" AND process.command_line: (*bypass* OR *hidden* OR *iex* OR *downloadstring* OR *encodedcommand*)`
Runs Every	5 minutes
Additional Look-back	6 minutes
Severity	High
Risk Score	73
MITRE ATT&CK	T1059.001 – PowerShell

Purpose

This rule searches for PowerShell process events containing command-line patterns that may be associated with suspicious execution, encoded commands, hidden execution, or download activity.

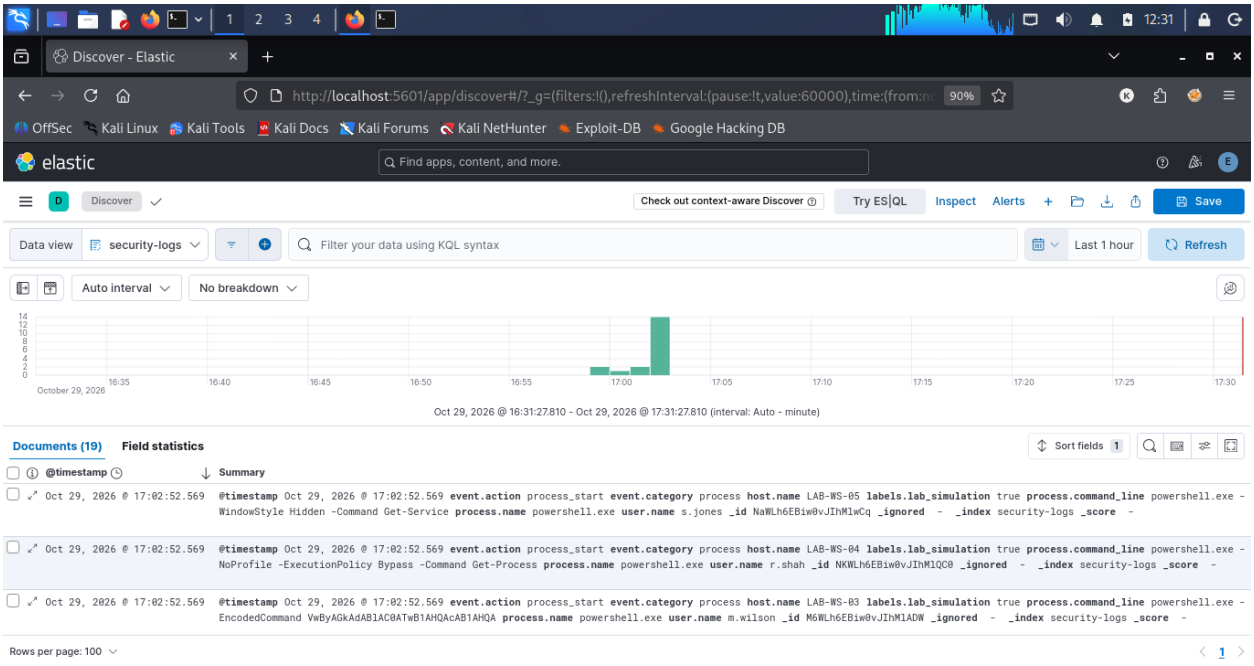
Testing and Expected Result

I used a controlled PowerShell test event to validate the query against the indexed logs. Matching events should be investigated by reviewing the complete command line, parent process, user account, endpoint, and related network activity.

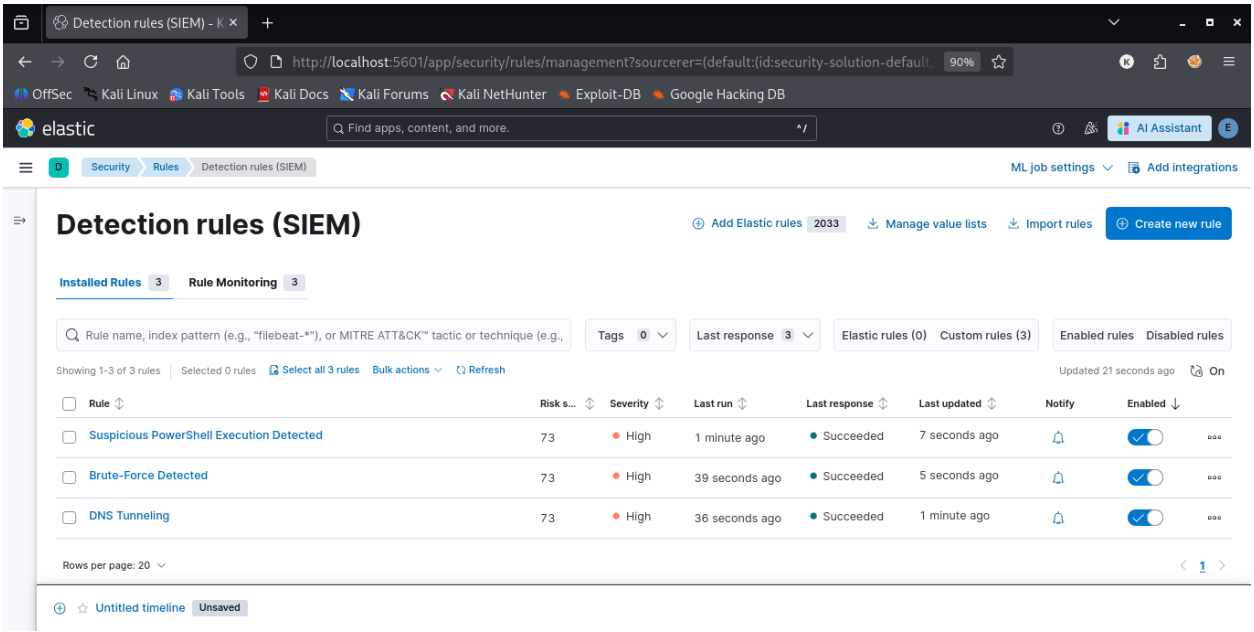
These command-line patterns are investigation indicators and may also appear in legitimate administrative activity.

7. Evidence and Screenshots

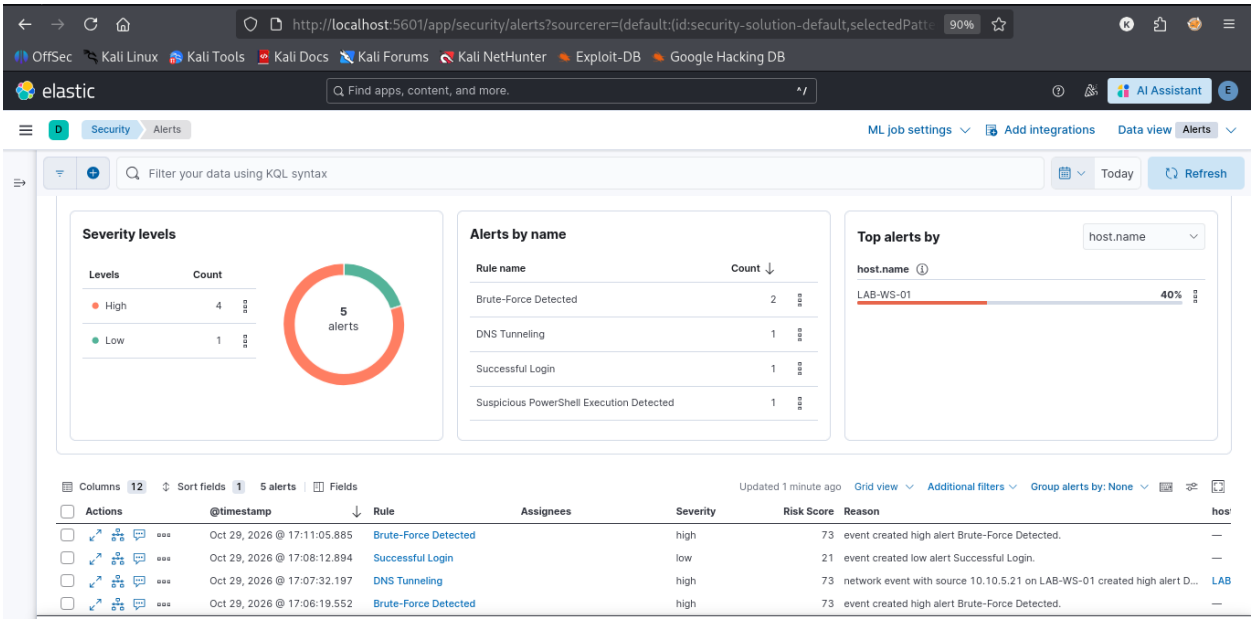
Screenshot 1: Security Logs Index



Screenshot 2: Detection Rules



Screenshot 3: Generated Alerts



8. Conclusion

I worked on configuring and testing three custom SIEM detection rules using the ELK Stack. The rules were designed to identify repeated authentication failures, potentially suspicious DNS activity, and suspicious PowerShell execution.

This task strengthened my practical understanding of security log analysis, detection engineering, KQL, and SIEM alert investigation. The final validation results are documented using screenshots from my own lab environment.